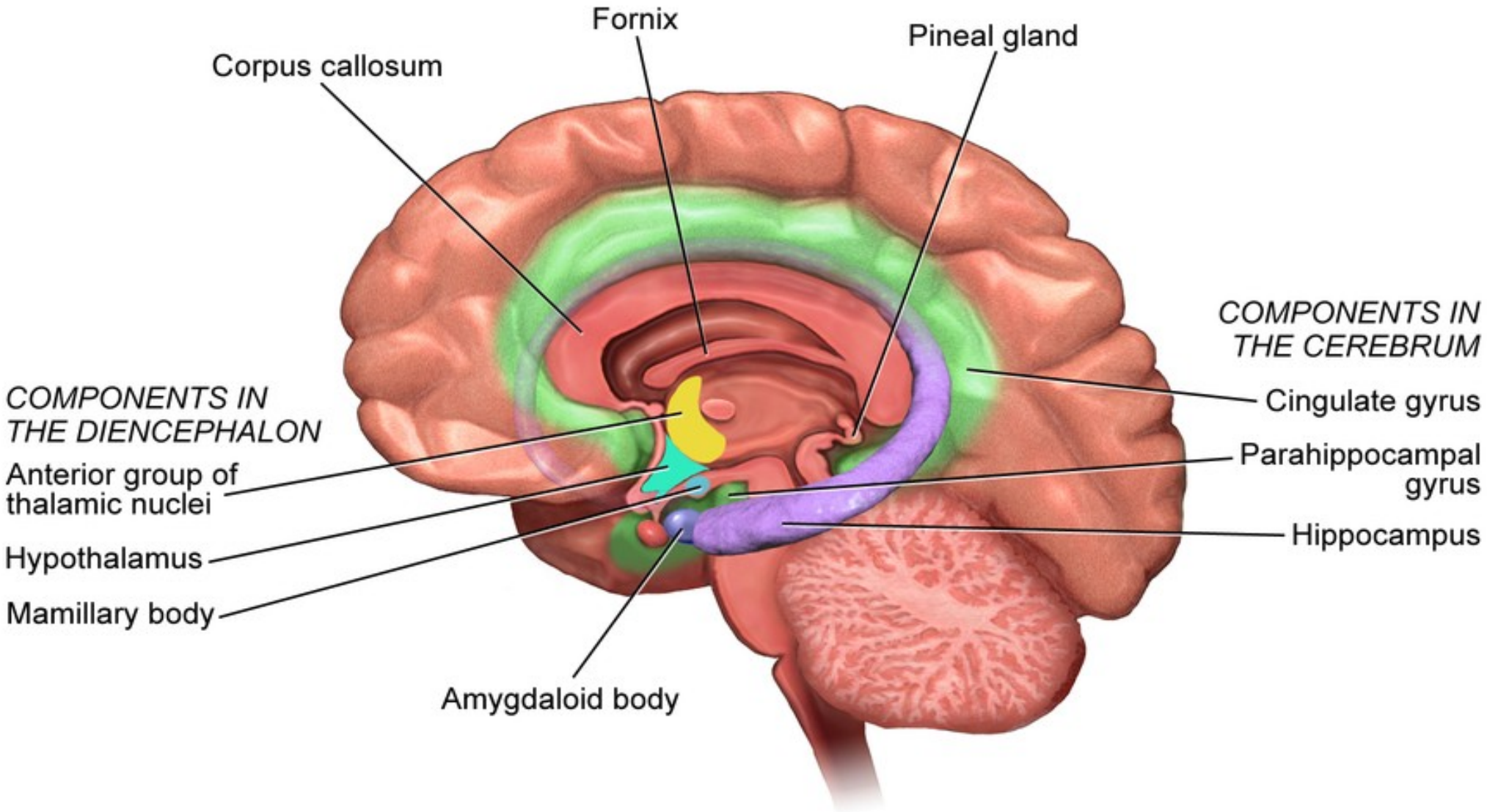
Limbic System

The Limbic System



- The limbic system is involved in essential survival behaviours, such as feeding, reproduction, caring for offspring and our fight or flight response.

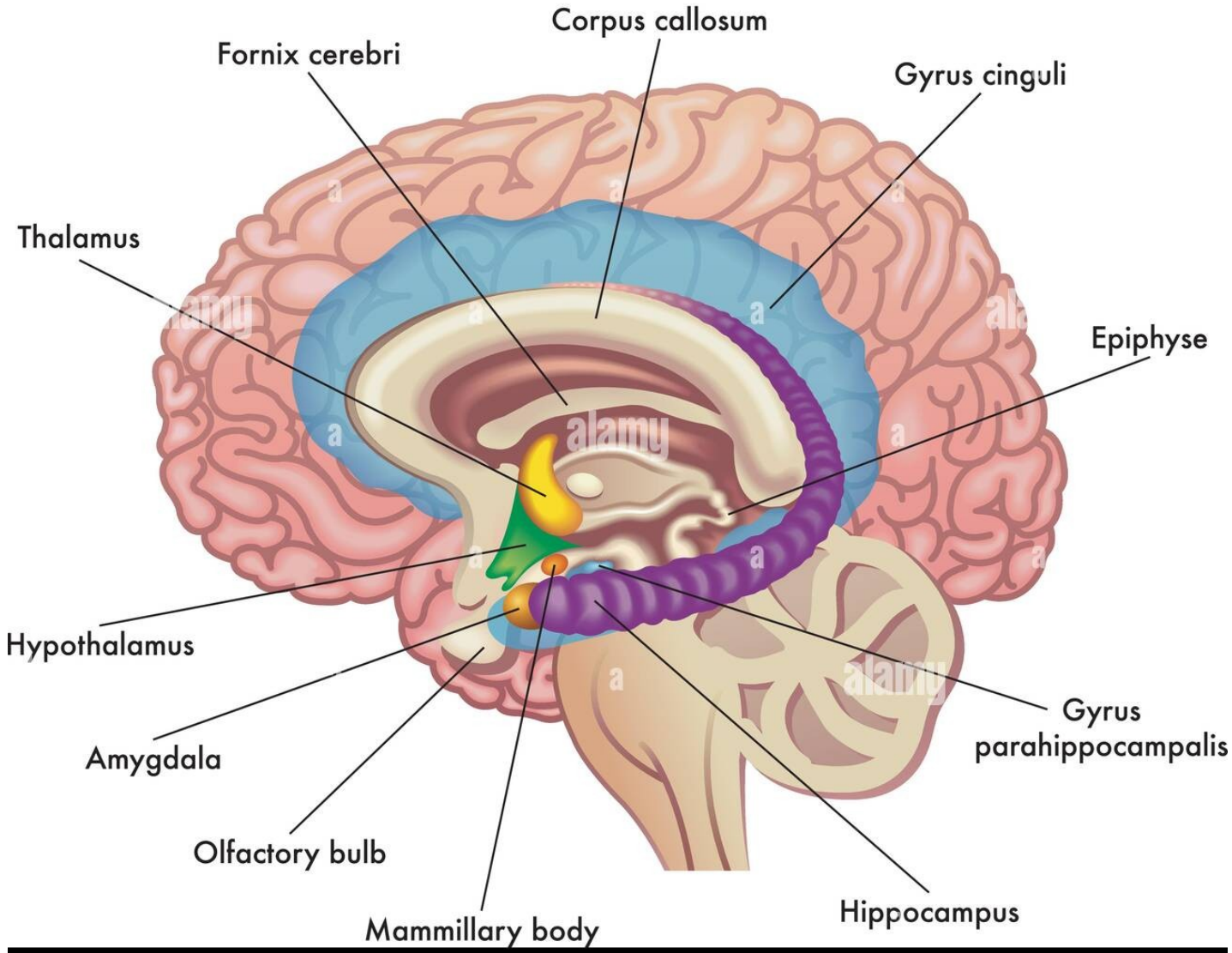
- t is a complex network of brain structures that are located lateral to the thalamus, beneath the cerebral cortex, and above the brainstem.

- In evolutionary terms, it is one of the oldest parts of the brain and it "constitutes a functional concept, and is characterized by dense afferent projections from brainstem and forebrain nuclei, which contributes to behavioral modulation

Important Limbic Structures

- amygdala,
- hippocampus,
- thalamus,
- hypothalamus,
- basal ganglia,
- cingulate gyrus

THE LIMBIC SYSTEM



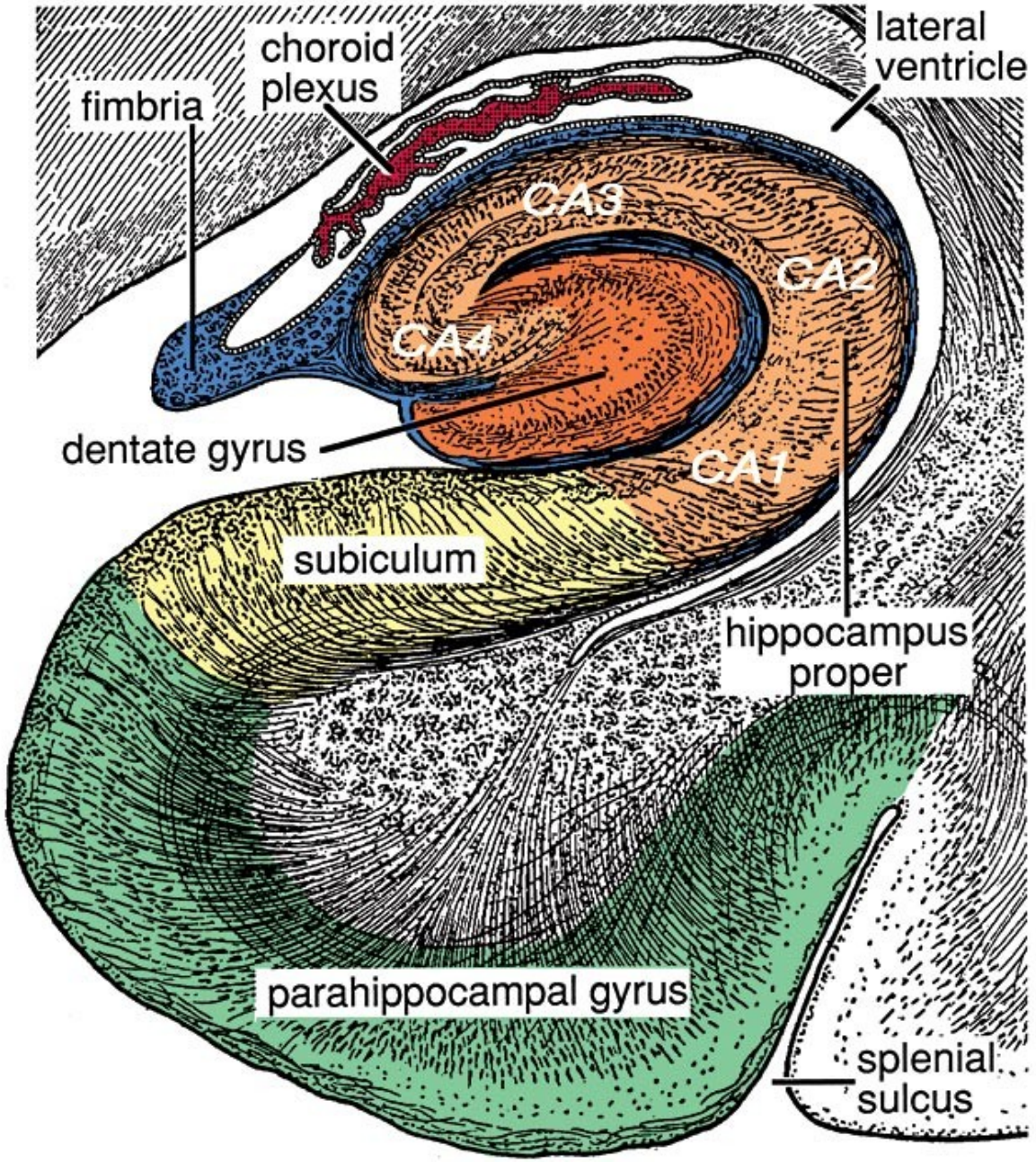
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Amygdala

- The amygdala is a small almond-shaped structure located in each temporal lobe, just below the uncus. We have two amygdalae, one in each hemisphere of the brain:
- The amygdalae play an important role in processing information between the prefrontal-temporal association cortices and the hypothalamus
- They are involved in regulating anxiety, aggression, fear conditioning, emotional memory and social cognition, behavioural and physiological stress responses

Hippocampus

- It has three regions: (1) dentate gyrus, (2) hippocampus proper, (3) subiculum:



- the hippocampus plays an essential role in forming *new* memories, particularly consolidated short- and long-term memories, spatial navigation, regulating emotions, motivation, hormones and autonomic activity
- with the prefrontal cortex, the hippocampus is involved in *episodic* memory
- the hippocampus is usually one of the first structures in the brain to be affected in Alzheimer's disease, and individuals with early Alzheimer's will have more difficulty remembering something they did that day compared to something they did as a child

Thalamus and Hypothalamus

- The **thalamus** functions as a relay station between the brain and the body, filtering through various types of sensory and motor information.
- The **hypothalamus** plays a major role in maintaining homeostasis via the autonomic nervous system, the neuroendocrine system, and the limbic system.

Cingulate Gyrus

- The cingulate gyrus is a paired structure that is located on the medial surface of each cerebral hemisphere, wrapping around the corpus callosum. It has four regions (1) anterior cingulate cortex (ACC) (2) midcingulate cortex, (3) posterior cingulate cortex (PCC) and (4) retrosplenial cortex.

Key functions

- processing emotions and regulating emotional responses
- cognitive processing (particularly reward-based decision-making)
- processing information from internal and external states to help guide motor responses
- imagination
- episodic memory formation and consolidation

Basal Ganglia

- The **basal ganglia** are a group of subcortical nuclei that are involved in motor control and learning, executive functions, emotional behaviours, reward and reinforcement, addictive behaviours and habit formation

Limbic System Disorders

- Potential signs or symptoms of limbic system disorders include: disinhibited behaviour
- difficulty regulating emotions
- hyperarousal
- appetite dysregulation
- trouble forming memories